

Learning Module: RM in openEHR, in details as a first introductory lesson

Section 1: Essential Vocabulary

Reference Model (RM) The set of fixed rules and definitions for information in openEHR.

EHR:	Electronic Health Record; a digital file of a patient's health.
Class:	A group or category in a computer model that defines how data looks.
Composition:	A document in the health record, like a doctor's note or a lab report.
Entry:	A specific piece of clinical information inside a document.
Observation:	Information about what we see, hear, or measure (like blood pressure).
Evaluation:	The doctor's opinion or diagnosis based on the facts.
Instruction:	An order to do something in the future, like giving medicine.
Action:	A record of what was actually done for the patient.
Attribute:	A specific detail or property of a class, such as a name or a date.

Section 2: Information Article

Introduction to the openEHR Reference Model (RM)

openEHR is a technology for storing health data. It helps hospitals and doctors share information easily. To do this, openEHR uses two main parts. The first part is the Reference Model, or RM. The second part is the Archetype Model, or AM. In this lesson, we will focus only on the Reference Model (RM).

What is the Reference Model?

The Reference Model (RM) is the foundation of openEHR. Think of it like the grammar of a language. It provides the rules for how we build sentences. In openEHR, the RM provides the rules for how we build health records. It defines the 'classes' and 'attributes' that software developers use. These rules do not change often. They are stable. This stability allows different computer systems to understand each other.

The Structure of the RM

The RM is organized in a hierarchy. This means there are big parts, and inside the big parts, there are smaller parts. Let us look at the main levels.

1. The EHR

At the very top is the EHR class. This represents the entire health record for one patient. It holds all the documents and folders for that person.

2. Composition

Inside the EHR, we have Compositions. A Composition is like a single document. It could be a 'Visit Note', a 'Discharge Summary', or a 'Lab Report'. Every piece of data in openEHR must belong to a Composition. A Composition always has a 'context'. The context tells us who wrote the document and when it happened.

3. Section

Inside a Composition, we can use Sections. A Section is like a heading in a book. It helps to organize the information. For example, a document might have a section called 'Family History' or 'Current Medicine'. Sections are useful, but they are not mandatory. You can put data directly into a

Composition without a Section.

4. Entry

The most important part of the RM is the Entry. The Entry is where the real clinical content lives. It is the statement about the patient. There are four main types of Entries in the RM:

- Observation: This is for information collected through seeing, hearing, or measuring. Examples are blood pressure, temperature, or the patient saying, 'I have a headache.' It is about facts.
- Evaluation: This is for the doctor's thinking. It includes the diagnosis or an assessment. It is an opinion based on the observations.
- Instruction: This is an order. It tells a nurse or another doctor to do something in the future. For example, 'Give 5mg of aspirin.'
- Action: This records what actually happened. If the nurse gives the aspirin, they record an Action. It tracks the progress of an Instruction.

Data Types

Finally, at the bottom of the structure, we have Data Types. These are the small details. The RM defines how to write text, numbers, and dates. For example, 'DvText' is for plain words. 'DvQuantity' is for numbers with units, like '50 kg'.

Conclusion

Understanding the RM is the first step to learning openEHR. It gives us the basic blocks: EHR, Composition, Section, and Entry. When we use these blocks correctly, we create safe and clear health records.

Section 3: Reading Comprehension Quiz

1. What does RM stand for?

Difficulty: EASY

- A) Real Model
- B) Reference Model
- C) Record Method
- D) Reference Method

2. Which part of openEHR defines the stable rules?

Difficulty: EASY

- A) The AM
- B) The RM
- C) The Hospital
- D) The Software

3. What is the top class in the hierarchy?

Difficulty: EASY

- A) Entry
- B) Section
- C) Composition
- D) EHR

4. What is a 'Composition' compared to?

Difficulty: EASY

- A) A single document
- B) A whole library
- C) A computer screen
- D) A single number

5. Which data type is used for plain words?

Difficulty: EASY

- A) DvQuantity
- B) DvText
- C) DvNumber
- D) DvDate

6. Based on the text, why are Sections described as 'optional'?

Difficulty: INTERMEDIATE

- A) Because they are expensive
- B) Because you can put data directly into a Composition
- C) Because they are not part of the RM
- D) Because doctors do not like them

7. If a doctor writes 'I think the patient has asthma', which Entry type is this?

Difficulty: INTERMEDIATE

- A) Observation (Fact)
- B) Evaluation (Opinion)
- C) Instruction (Order)
- D) Action (Done)

8. What is the relationship between Instruction and Action?

Difficulty: INTERMEDIATE

- A) Action happens before Instruction
- B) Instruction tracks the progress of Action
- C) Action tracks the progress of Instruction
- D) They are the same thing

9. Why does the text compare the RM to grammar?

Difficulty: INTERMEDIATE

- A) Because it is hard to learn
- B) Because it provides rules for building the record
- C) Because it uses English words
- D) Because it is only for writing

10. What is the main purpose of using the RM according to the conclusion?

Difficulty: INTERMEDIATE

- A) To save paper
- B) To create safe and clear health records
- C) To make doctors work faster
- D) To stop using computers

Section 4: Reading Quiz Answer Key

Question 1: Reference Model Question 2: The RM

Question 3: EHR Question 4: A single document

Question 5: DVText Question 6: Because you can put data directly into a Composition

Question 7: Evaluation (Opinion) Question 8: Action tracks the progress of Instruction

Question 9: Because it provides rules for building the record Question 10: To create safe and clear health records

Section 5: Dialogue Transcript (Listening)

- Jane:** Hello, Joe. How are you today? Are you ready to start our lesson?
- Joe:** Hi Jane. I am good, thank you. Yes, I am ready. I want to learn about openEHR.
- Jane:** That is great. Today, we will talk about the Reference Model. We often call it the RM. Have you heard of the RM before?
- Joe:** I have heard the name, but I do not know what it means. Is it a computer program?
- Jane:** Not exactly. It is not a program you run. It is a set of rules. It is a model for information. Think of it like a blueprint for a house.
- Joe:** A blueprint. Okay. So it tells us how to build the health record?
- Jane:** Yes, exactly. It defines the structure. In openEHR, we have two levels. One is the RM, and the other is the Archetypes. Today, we only look at the RM.
- Joe:** Okay. Only the RM. What is the most important thing in the RM?
- Jane:** The RM has many classes. A class is a group of data. The biggest class is the EHR. EHR stands for Electronic Health Record.
- Joe:** I see. So the EHR class is the whole file for the patient?
- Jane:** Yes. Every patient has one EHR. Inside the EHR, we have documents. In the RM, we call a document a 'Composition'.
- Joe:** Composition. That is a new word for me. Can you give me an example of a Composition?
- Jane:** Sure. When you go to the doctor, the doctor writes notes. That note is a Composition. If you get a blood test, the result paper is a Composition.
- Joe:** I understand. A Composition is one event or one report. What is inside the Composition?
- Jane:** Good question. A Composition contains 'Content'. We can organize this content using 'Sections'.
- Joe:** Sections? Like chapters in a book?
- Jane:** Yes, exactly like chapters. You might have a Section for 'Allergies' and a Section for 'Medications'. But Sections are optional. You do not always need them.
- Joe:** Okay. If I do not use a Section, what do I use? What holds the actual data?
- Jane:** The actual clinical data lives in the 'Entry' classes. The Entry is the most important part of the RM for clinical details.
- Joe:** Entry. Okay. Are there different types of Entries?

Jane: Yes, there are. There are four main types of Entries in the RM. They are Observation, Evaluation, Instruction, and Action.

Joe: That sounds detailed. Can we talk about them one by one? What is an Observation?

Jane: Of course. An Observation is information that you can see, hear, or measure. It is objective. It is a fact about the present or the past.

Joe: Can you give me an example of an Observation?

Jane: Yes. Imagine a nurse measures your temperature. The number on the thermometer is an Observation. If you tell the doctor, 'My leg hurts,' that is also an Observation. It is a symptom.

Joe: I see. So, lab results are Observations too?

Jane: Correct. A blood test result is an Observation. Now, the second type is Evaluation. Do you know what evaluation means in general English?

Joe: It means to judge something, or to give an opinion.

Jane: Perfect. In the openEHR RM, an Evaluation is the doctor's opinion. It is a diagnosis. It is what the doctor thinks about the Observations.

Joe: So, if the Observation is 'high temperature', the Evaluation is 'You have the flu'?

Jane: That is a perfect example, Joe! The temperature is the fact. The flu diagnosis is the assessment. That is the Evaluation.

Joe: Okay, that is clear. What is the third type?

Jane: The third type is Instruction. An Instruction is when the doctor tells someone to do something. It is about the future.

Joe: Is it like a prescription?

Jane: Yes. A prescription for medicine is an Instruction. A referral to another doctor is an Instruction. It is a plan.

Joe: Okay. And the last one is Action?

Jane: Yes, Action. An Action records what was done. It is linked to the Instruction.

Joe: How is it linked?

Jane: Well, imagine the doctor creates an Instruction: 'Give patient an injection.' Later, the nurse gives the injection. The nurse records an Action: 'Injection given at 2 PM'.

Joe: I understand. The Instruction is the plan, and the Action is the reality.

Jane: Exactly. This helps us track the workflow. We know what was ordered and what was finished.

Joe: This structure seems very organized. Observation, Evaluation, Instruction, Action. Is there anything smaller than an Entry?

Jane: Yes. Inside the Entries, we need to store the specific values. We use 'Data Types' for this. The RM defines many Data Types.

Joe: What is a Data Type?

Jane: A Data Type tells the computer what kind of information it is holding. For example, is it a word? Is it a number? Is it a date?

Joe: Can you name some Data Types in openEHR?

Jane: Sure. One common type is 'DvText'. 'Dv' stands for Data Value. DvText is for plain text.

Joe: Like a comment?

Jane: Yes, like 'Patient looks tired'. That is DvText. Another important one is 'DvQuantity'. This is for numbers with units.

Joe: So, '80 kg' uses DvQuantity?

Jane: Yes. The number is 80, and the unit is kg. DvQuantity keeps them together. We also have 'DvDate' and 'DvTime'.

Joe: That makes sense. Why are these rules so strict? Why can we not just write everything in a Word document?

Jane: If we use a Word document, the computer cannot understand the data. It just sees letters. With the RM, the computer knows that this number is a weight, and that text is a diagnosis.

Joe: So it helps computers talk to each other?

Jane: Yes, that is called interoperability. The RM makes interoperability possible because everyone follows the same rules.

Joe: I think I have a good picture now. We have the EHR. Inside is the Composition. Inside that are Sections and Entries. And Entries have Data Types.

Jane: You summarized it perfectly, Joe. You are learning fast.

Joe: Thank you, Jane. Is there anything else about the Composition I should know?

Jane: Yes, one small detail. Every Composition has a 'context'. The context is very important.

Joe: What does the context tell us?

Jane: The context tells us the setting. It answers questions like: Who is the patient? Who is the doctor? Where did this happen? When did it happen?

Joe: So without context, we do not know whose data it is?

Jane: Exactly. The data might be correct, but without context, it is useless. The RM ensures every Composition has this context.

Joe: This is very interesting. The RM seems like a very strong foundation.

Jane: It is. It has been designed by many experts over many years. It covers almost every situation in healthcare.

Joe: What if I need to record something very strange? Does the RM handle it?

Jane: Usually, yes. Because the RM classes like Observation and Evaluation are generic. You can put any specific medical content inside them using Archetypes.

Joe: Ah, Archetypes. You mentioned them at the beginning. But we are not studying them today, right?

Jane: That is right. Today is just for the RM. But remember, the RM is the skeleton. The Archetypes are the muscles and skin.

Joe: That is a good analogy. Skeleton and muscles. I will remember that.

Jane: Do you have any more questions about the four types of Entries?

Joe: Let me think. We talked about Observation, Evaluation, Instruction, and Action. What if I want to record a family history? My father had heart disease.

Jane: That is a good question. Family history is usually an Evaluation.

Joe: Why is it an Evaluation? Isn't it a fact?

Jane: It is a fact that your father had it. But for your health record, it is a risk assessment. It summarizes the risk to you. However, sometimes people model it as an Observation of the family. It depends on how you design the Archetype.

Joe: I see. So sometimes it can be tricky?

Jane: Yes, sometimes. But usually, the difference is clear. Observation is raw data. Evaluation is synthesis or opinion.

Joe: Okay. And what about a survey? If I answer questions on a paper.

Jane: Your answers to a survey are Observations. You are observing your own feelings or status.

Joe: Okay. And if the doctor reads my survey and says I am depressed?

Jane: Then the doctor's statement is an Evaluation. You observe your feelings. The doctor evaluates your condition.

Joe: That makes it very clear. Thank you, Jane.

Jane: You are welcome, Joe. We have covered a lot of ground today. We discussed the EHR, Compositions, Sections, Entries, and Data Types.

Joe: Yes, and the Context too.

Jane: Yes. Do you feel confident about the basics of the openEHR RM now?

Joe: Yes, I do. I feel ready to read more about it.

Jane: Excellent. Next time, we can look at how to build an Archetype. But for now, please review these notes.

Joe: I will review them tonight. Thanks for the lesson, Jane.

Jane: My pleasure. Have a good evening, Joe.

Joe: You too. Goodbye.

Section 6: Listening Comprehension Quiz

1. What topic does Joe want to learn about?

- A) How to build a house
- B) The openEHR Reference Model
- C) How to be a doctor
- D) The Archetype Model

2. How does Jane describe the Reference Model (RM)?

- A) It is a computer program you run
- B) It is a set of rules or a blueprint
- C) It is a type of medicine
- D) It is a hospital building

3. What is the biggest class in the RM?

- A) Composition
- B) Entry
- C) EHR
- D) Section

4. What represents a single document in the RM?

- A) Composition
- B) EHR
- C) Folder
- D) Observation

5. Are Sections mandatory (required) in a Composition?

- A) Yes, always
- B) No, they are optional
- C) Only in lab reports
- D) Yes, but only one

6. Where does the real clinical data live?

- A) In the EHR class
- B) In the Section class
- C) In the Entry class
- D) In the Context

7. Which Entry type is for things you see, hear, or measure?

- A) Evaluation
- B) Instruction
- C) Action
- D) Observation

8. According to Jane, a diagnosis is which type of Entry?

- A) Observation
- B) Evaluation
- C) Instruction
- D) Action

9. What is an example of an Instruction?

- A) A blood pressure reading
- B) A diagnosis of flu
- C) A prescription for medicine
- D) A record of a past surgery

10. What does an Action record?

- A) What was actually done
- B) What the doctor thinks
- C) What might happen
- D) The patient's name

11. Which Data Type is used for numbers with units (like 80 kg)?

- A) DvText
- B) DvDate
- C) DvQuantity
- D) DvTime

12. Why is the RM important for computers?

- A) It makes them faster
- B) It helps them understand the data (interoperability)
- C) It makes them cheaper
- D) It prevents viruses

13. What does the 'context' of a Composition tell us?

- A) The price of the medicine
- B) Who, when, and where the document was created
- C) The list of all patients
- D) The future plan

14. In Jane's analogy, if the RM is the skeleton, what are the Archetypes?

- A) The brain
- B) The clothes
- C) The muscles and skin
- D) The shoes

15. How does Jane classify a patient survey?

- A) As an Instruction
- B) As an Observation
- C) As an Action
- D) As a Composition

Section 7: Listening Quiz Answer Key

Q1: The openEHR Reference Model | Q2: It is a set of rules or a blueprint | Q3: EHR

Q4: Composition | Q5: No, they are optional | Q6: In the Entry class

Q7: Observation | Q8: Evaluation | Q9: A prescription for medicine

Q10: What was actually done | Q11: DvQuantity | Q12: It helps them understand the data (interoperability)

Q13: Who, when, and where the document was created | Q14: The muscles and skin | Q15: As an Observation

Teacher's Overall Feedback:

The openEHR Reference Model It is a set of rules or a blueprint EHR Composition No, they are optional In the Entry class Observation Evaluation A prescription for medicine What was actually done DvQuantity It helps them understand the data (interoperability) Who, when, and where the document was created The muscles and skin As an Observation Reference Model The RM EHR A single document DvText Because you can put data directly into a Composition Evaluation (Opinion) Action tracks the progress of Instruction Because it provides rules for building the record To create safe and clear health records